

Le Huu Huy

 github.com/lhhuy02012005  [Portfolio](#)  hhuy.2005@gmail.com

EDUCATION

Sai Gon University <i>B.S. Computer Science</i>	Step.2023 - June 2027 <i>Current GPA: 3.16/4.0</i>
---	---

SKILLS

Languages: Java, PHP, Python, JavaScript/TypeScript, C/C++, C#, HTML/CSS
Frameworks & Libraries: Spring Boot, Spring Data JPA, Hibernate, MapStruct, Laravel, React 19, FastAPI, LangChain
Security & IAM: Spring Security, FusionAuth, OAuth2 (Google Login), JWT
Databases & Caching: PostgreSQL, MySQL, Redis, ChromaDB, NoSQL
Architecture & Tools: Docker, Git/GitHub, Unix Shell, Maven, Postman, WebSocket, Cloudinary

PROJECTS

ShopZues — Scalable E-commerce Platform <i>Laravel, React, Firebase, MySQL, Redis</i>	Nov 2023
- Implemented a scalable Laravel backend (PHP 8.4) with JWT auth, RBAC (role/permission), and Redis (cache + queue + OTP). - Integrated Firebase Realtime DB for live order status & notifications, and Brevo for OTP email verification. - Developed React frontend + Flutter mobile app that connect to backend APIs and cloud DB (Firebase + MySQL). - Used Docker + docker-compose for consistent dev/test environments and easy deployment. - Learned end-to-end full-stack architecture: auth flows, real-time sync, payment flow, and micro-service patterns.	

RAGAI (NexusRAG) — Hybrid Knowledge Base & Agentic Chat	2026
Tech Stack: <i>FastAPI, React 19, TypeScript, PostgreSQL, ChromaDB, LightRAG, Ollama, Sentence-Transformers, Docker</i> - Designed an end-to-end local-first RAG architecture: document upload → Docling parsing → chunking/dedup → embedding → vector + knowledge-graph indexing → hybrid retrieval → grounded generation. - Implemented a hybrid retrieval pipeline combining ChromaDB vector search and LightRAG graph context, then improved precision with cross-encoder reranking (BAAI/bge-reranker-v2-m3). - Built agentic chat with tool-calling (<code>search_documents</code>) and SSE streaming, delivering real-time events for thinking, tokens, sources, images, and final answer. - Developed citation-first response format with source traceability (document/page/heading path), enabling explainable answers and quick navigation back to evidence. - Added visual document intelligence: extracted images/tables from PDFs, generated captions, and injected media semantics into chunk embeddings for better retrieval quality. - Engineered robust document lifecycle handling (PARSING/INDEXING/INDEXED/FAILED), timeout recovery, and async background processing for long-running ingestion tasks. - Implemented workspace isolation across PostgreSQL metadata, vector collections, and knowledge-graph storage, supporting multi-tenant knowledge bases safely. - Containerized infrastructure with Docker Compose (PostgreSQL + ChromaDB) and tuned runtime configs (chunk size, prefetch, rerank top-k, KG extraction limits) for both medium and high-spec machines.	

EXPERIENCE

Sai Gon University

Software Engineering Student

Ho Chi Minh City, Viet Nam

Sept. 2023 – Present

- Developed high-performance web systems using **Laravel** and **Spring Boot**, focusing on Service-Repository patterns and clean architecture.
- Architected secure authentication systems using **JWT**, **OAuth 2.0**, and **Spring Security**, managing stateless sessions and role-based access control (RBAC).
- Optimized application performance by implementing **Redis** for distributed caching and asynchronous background tasks (queues), reducing database load during high-traffic simulations.
- Integrated third-party ecosystems including **VNPAY** for payment processing, **GHN** for logistics automation, and **Firebase** for real-time data synchronization.
- Containerized applications with **Docker** and orchestrated development environments using Docker Compose to ensure cross-platform consistency.
- Collaborated on version control via **Git/GitHub**, managing complex branching workflows and maintaining codebase integrity through code reviews.

HOBBIES

Music & Arts: **Playing badminton** , **Gym** , **Swimming**.

Tech Exploration: Researching **Generative AI (RAG)** and local LLM deployment; building custom mechanical keyboards.